College of Engineering Trivandrum

Department of Computer Applications

20MCA203 . Design and Analysis of Algorithms (Semester 3)

Tutorial I

1. Consider the following code:

```
for (i = 0; i < n; i++)
   for (j = 0; j < i; j++)
   {
      stmt;
   }
```

   Determine the asymptotic time complexity of the above code. Show the number of times each loop is executed and the total number of executions.

2. Consider the following code:

```
p = 0;

for (i = 1; p <= n; i++)

        p = p + i;
```

   Determine the asymptotic time complexity of the above code. Show the number of times each loop is executed and the total number of executions.

3. Consider the algorithm:

```
 if (n > 5)

         printf("%d", n);

else

        for (i = 0; i < n; i++)

                printf("%d", i);
```

   Determine the asymptotic time complexity. Show the total number of steps

4. Consider the code fragment:

```
a = 1;

while (a < n)

{       Stmt;

        a = a * 2; }
```

Determine the asymptotic time complexity. Show the number of times each step is executed

5.
```
for (i = 0; i < n; i++)

        for (j = 1; j < n; j = j * 2)

                {  stmt;   }
```

Determine the asymptotic time complexity of the above code. Show the number of times each loop is executed and the total number of executions.